SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Solar Inverter 25kW (PV-INV-SOLAR-25) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 500 mm above, 300 mm below and on each side; avoid direct sunlight on the heat sink
- Operating temperature: -25 to +60 °C (derating above 45 °C)
- Degree of protection: IP66
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Measure Voc of each string at the lowest expected temperature
2. Check DC polarity and insulation
3. Select grid code / country setting
4. Verify communication and monitoring
5. Check MPPT operation at noon

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Insulation fault at sunrise	Firmware defect at humidity > 95%	TG-PV-INV-SOLAR-25 §3.1
Insulation fault after rain	Water ingress in PV connector	TG-PV-INV-SOLAR-25 §3.2
Grid lost alarm	AC breaker tripped / grid outage	TG-PV-INV-SOLAR-25 §3.3
PV over-voltage alarm in winter	String too long for low temperature Voc	TG-PV-INV-SOLAR-25 §3.4

On-site commissioning support can be requested through the Global CS Center. For pv inverters the request is handled by the technical support team.